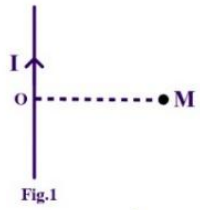
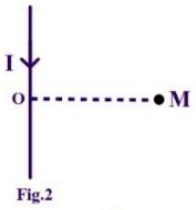
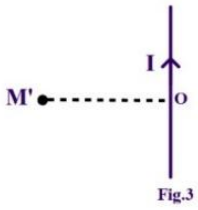
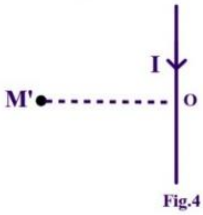
Chapter4 Lesson1 Electromagnet (Section1)

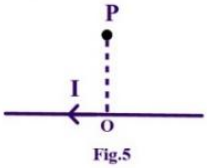
1. Use the answer key provided to indicate the direction of the magnetic field on the current carrying-wire at point **M**, **M'**, **P**, and **P'** in diagrams **(1)** to **(6)** below.

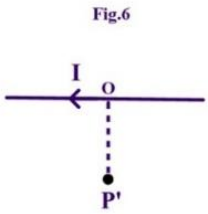


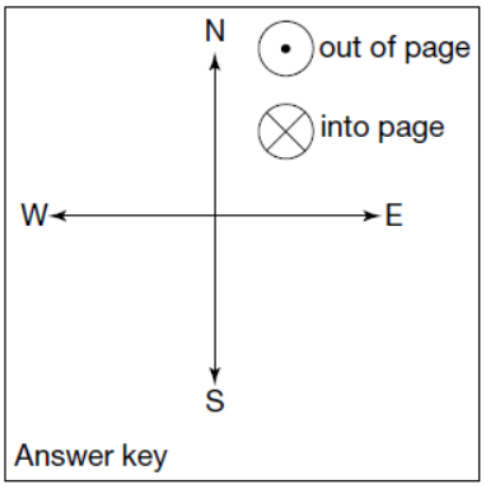




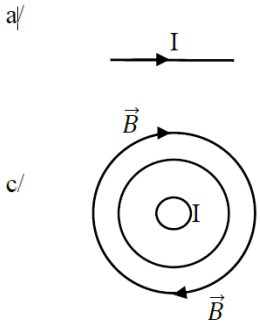


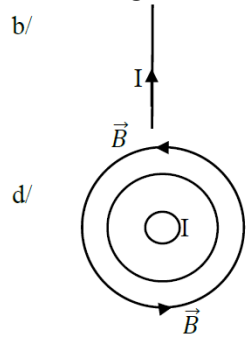


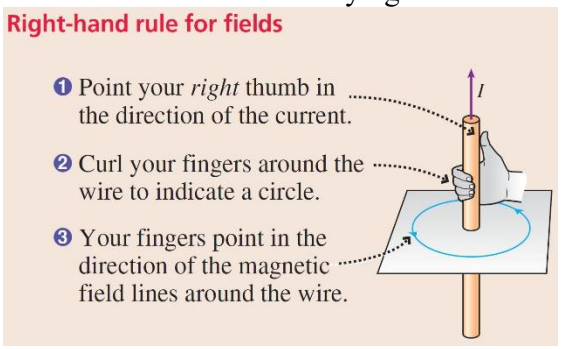




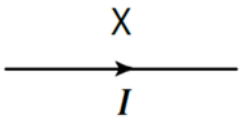
2. Use the right-hand rule to find the direction of magnetic field and electric current in current carrying wire below:

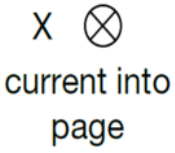


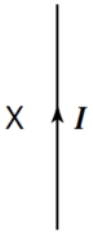


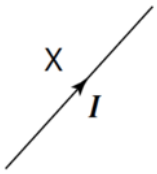


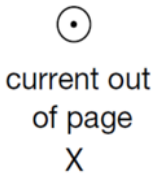
3. Use the right-hand rule to find the direction of magnetic field at point X in a current-carrying wire below:



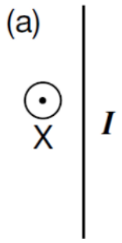


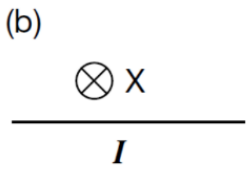


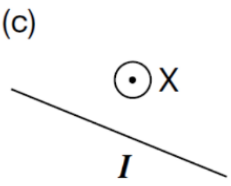


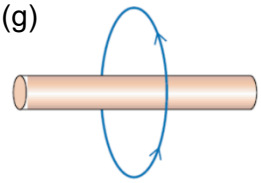


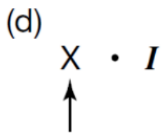
4. Use the right-hand rule and the direction of the magnetic field at X to determine the direction of the current in the wire in each case in diagrams (a) to (e) below and direction of the current in each wire in diagrams (f) to (h) below.

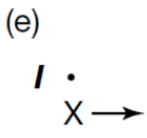


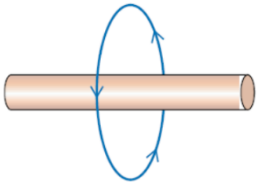


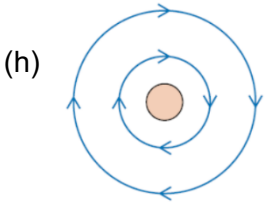












5. **Figure 6**: Indicate on each diagram the direction of the electric current, the magnetic field lines around the coil, and the north and south poles of the coil.

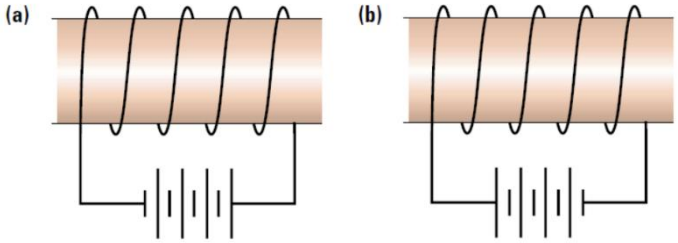
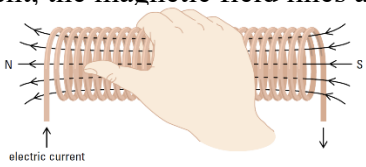
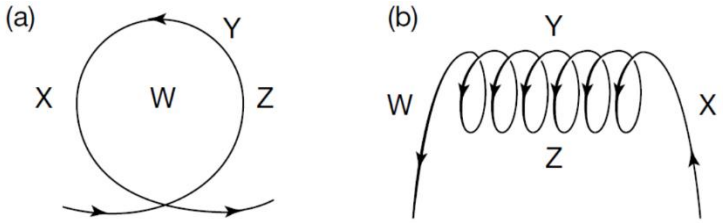


Figure 6

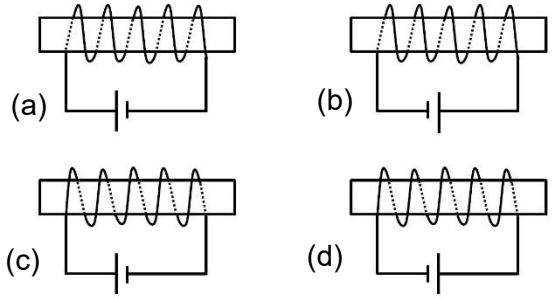


Right-Hand Rule Grip for a Solenoid
- Wrap your 4 fingers around the solenoid in the same way as electric current flow through the loop of wire.
- Outstretch your right thumb in the direction of magnetic field (N pole).

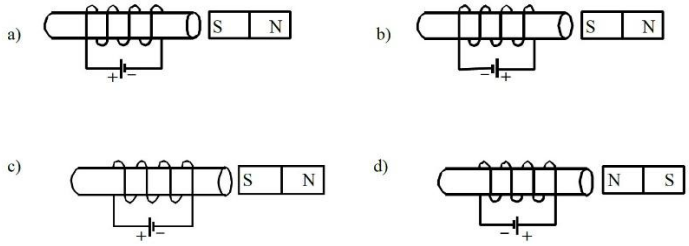
6. Use the right-hand-grip rule to determine the direction of the magnetic field at W, X, Y, Z in the following diagrams. Figure **(a)** represents a circular loop of wire with a current and figure **(b)** represents a solenoid.



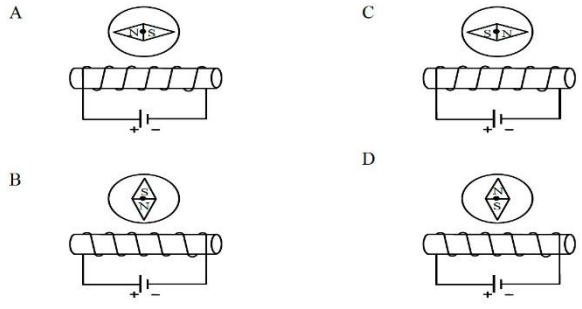
7. The four diagrams below represent electromagnets connected to the terminals of a battery. In each of the diagrams, indicate the magnetic poles of the electromagnet. Find the direction of magnetic field?



8. A bar magnet is brought close to a current-bearing electromagnet. Indicate whether there will be attraction or repulsion between the electromagnet and the bar magnet.

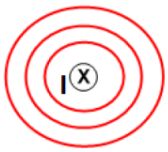


9. A compass is placed in a magnetic field. Which of the following diagrams shows the compass needle pointing in the correct direction?



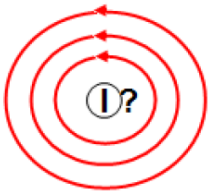
10. An electric current flow into the page. What is the direction of the magnetic field?

A. To the bottom of the page.
B. To the top of the page.
C. Clockwise.
D. Counter-clockwise.
E. To the right.



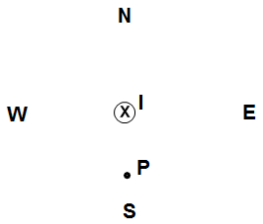
11. A current-carrying wire is placed perpendicular to the page. Determine the direction of the electric current from the direction of the magnetic field.

A. Into the page.
B. Out of the page.
C. Clockwise.
D. Counter-clockwise.
E. To the left.



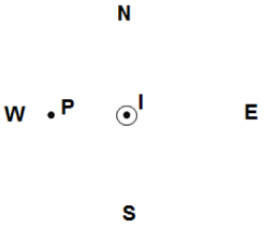
12. A vertical wire carries an electric current into the page. What is the direction of the magnetic field at point P located to the south from the wire?

A. West.
B. North.
C. East.
D. South.
E. Down.

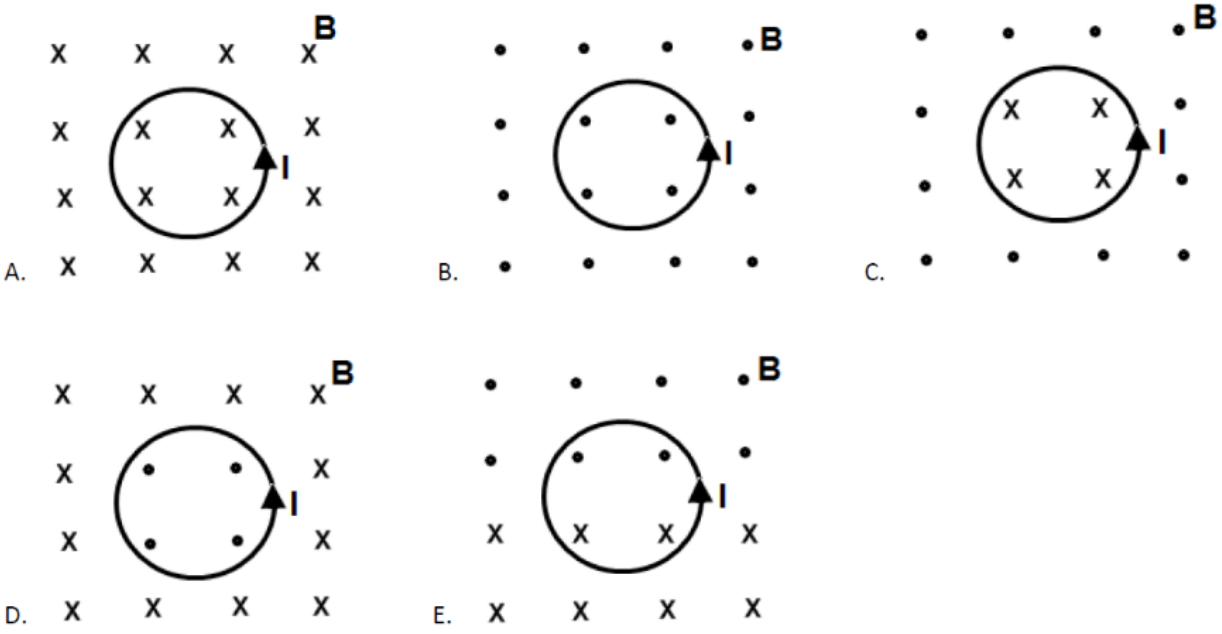


13. A vertical wire carries an electric current out of the page. What is the direction of the magnetic field at point P located to the west from the wire?

A. West.
B. North.
C. East.
D. South.
E. Down.



14. Which of the following diagrams represents the magnetic field due to a circular current?

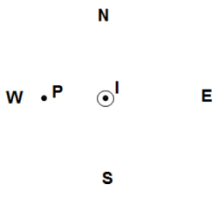


Chapter4 Lesson1 Electromagnet (Section2)

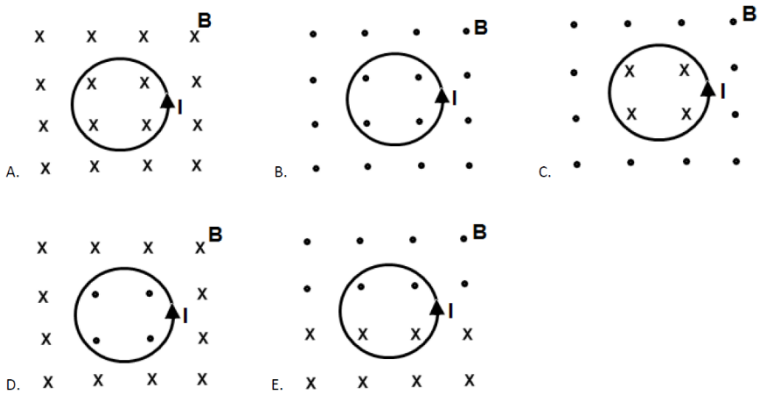
15. State the Fleming's Right Hand Rule.

16. A vertical wire carries an electric current out of the page. What is the direction of the magnetic field at point P located to the west from the wire?

A. West.
B. North.
C. East.
D. South.
E. Down.

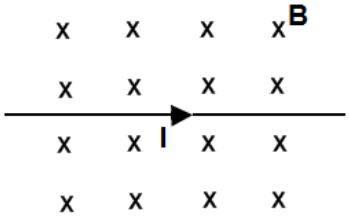


17. Which of the following diagrams represents the magnetic field due to a circular current?



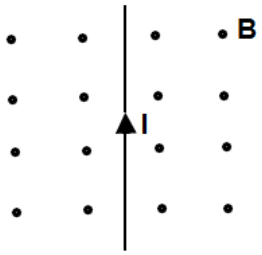
18. A straight long wire carries an electric current to the right. The current is placed in a uniform magnetic field directed into the page. What is the direction of the magnetic force on the current?

A. Left.
B. Right.
C. To the bottom of the page.
D. To the top of the page.
E. Out of the page.



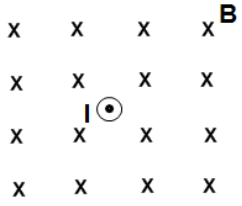
19. A straight long wire carries an electric current to the top of the page. The current is placed in a uniform magnetic field directed out the page. What is the direction of the magnetic force on the current?

A. Left.
B. Right.
C. To the bottom of the page.
D. To the top of the page.
E. Out of the page



20. A straight long wire carries an electric current out the page. The current is placed in a uniform magnetic field directed into the page. What is the direction of the magnetic force on the current?

A. Left.
B. Right.
C. To the bottom of the page.
D. To the top of the page.
E. There is no magnetic force on the current.

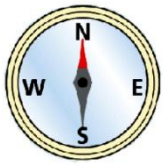


21. Copy and complete these sentences. You may need to use the same word or phrase more than once.
a. We can draw …… …… lines to show the pattern of the …… …… around a bar magnet.

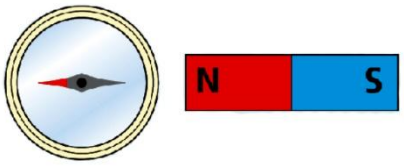
b. If there are lots of lines close together, that means that the …… …… is very ……... . This happens near …… of the magnet.

c. All magnets have a north and a south ……

22. Here is a diagram of a plotting compass when there is no magnet near it and when it is near the end of a bar magnet.



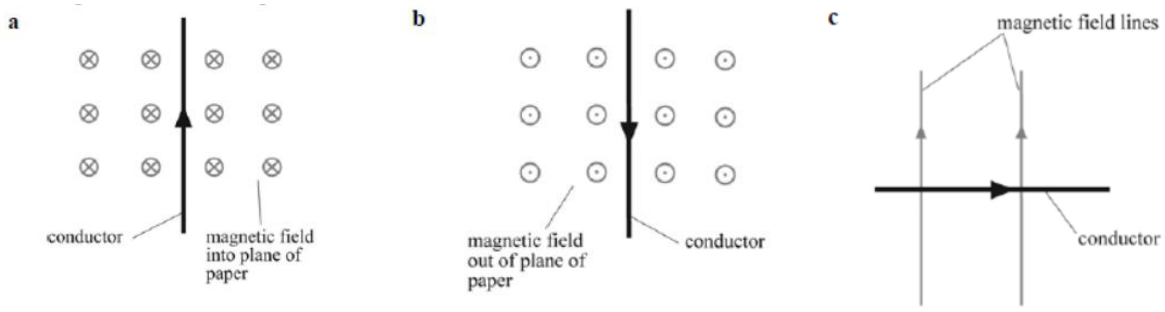
A compass with no magnet near it.



B compass near the end of a bar magnet.

Explain what is happening to the needle in each picture.

23. A current-carrying conductor is placed in an external magnetic field. In each case below, use Fleming's left-hand rule to predict the direction of the force on the conductor.

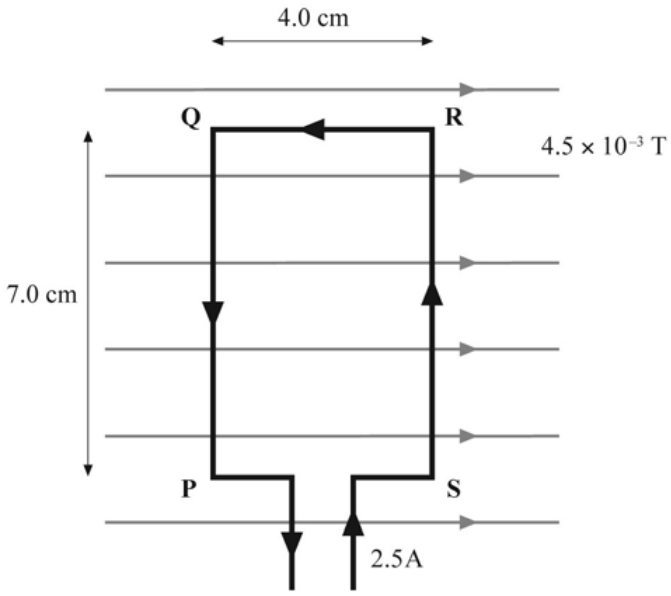


24. The diagram shows a rectangular metal frame **PQRS** placed in a uniform magnetic field. The magnetic flux density is 4.5×10^{-3} T. The current in the metal frame is 2.5 A.

a Calculate the force experienced by side **PQ** of the frame.

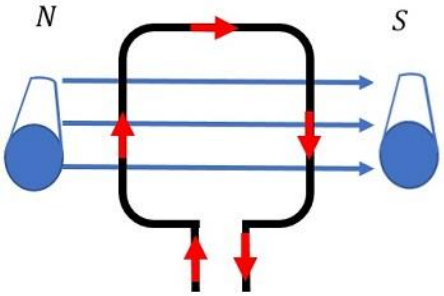
b Suggest why side **QR** does not experience a force.

c Describe the motion of the frame immediately after the current in the frame is switched on.



25. A current-carrying conductor placed at right angles to a uniform magnetic field, experiences a force of 4.70×10^{-3} N.
Determine the force on the wire when, separately:
a the current in the wire is increased by a factor of 3.0
b the magnetic flux density is halved
c the length of the wire in the magnetic field is reduced to 40% of its original length.

26. In the diagram, we are looking down on the setup. The current is moving clockwise around the coil. Which way does the coil rotate?



a. The right side goes down, and the left side goes up.
b. The right side goes up, and the left side goes down.
c. The whole coil moves up.
d. The whole coil moves down.

27. The device used for producing electric current is called a
(a) generator.
(b) galvanometer.
(c) ammeter.
(d) motor.

28. The essential difference between an AC generator and a DC generator is that
(a) AC generator has an electromagnet while a DC generator has permanent magnet.
(b) DC generator will generate a higher voltage.
(c) AC generator will generate a higher voltage.
(d) AC generator has slip rings while the DC generator has a commutator.

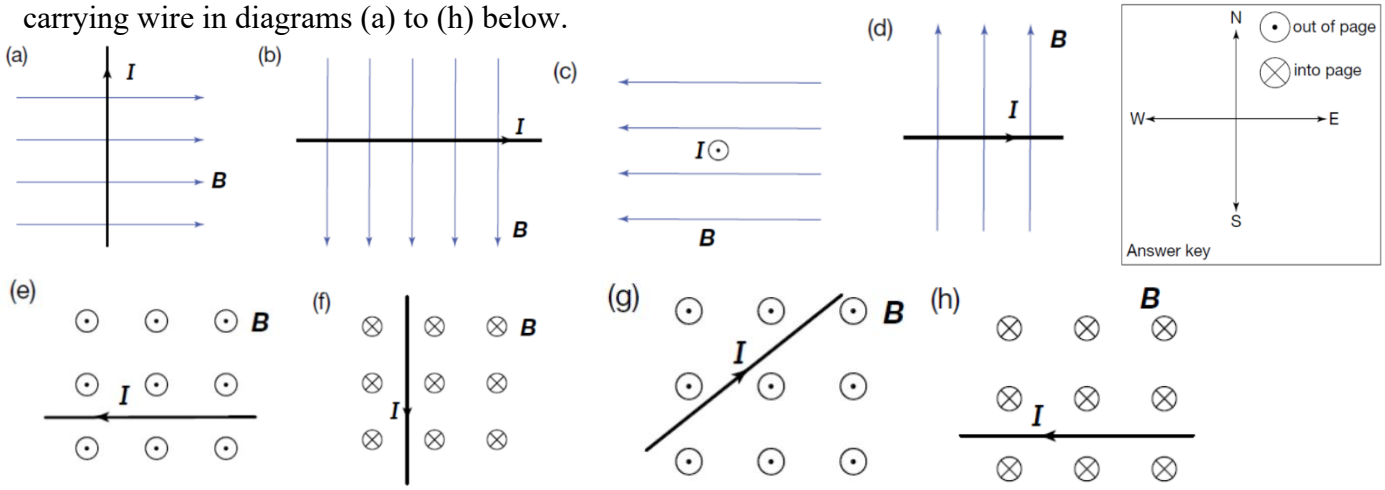
29. At the time of short circuit, the current in the circuit
(a) reduces substantially
(b) does not change
(c) increases heavily
(d) vary continuously

30. State whether the following statements are true or false.
(a) An electric motor converts mechanical energy into electrical energy.
(b) An electric generator works on the principle of electromagnetic induction.
(c) The field at the centre of a long circular coil carrying current will be parallel straight lines.
(d) A wire with a green insulation is usually the live wire of an electric supply.

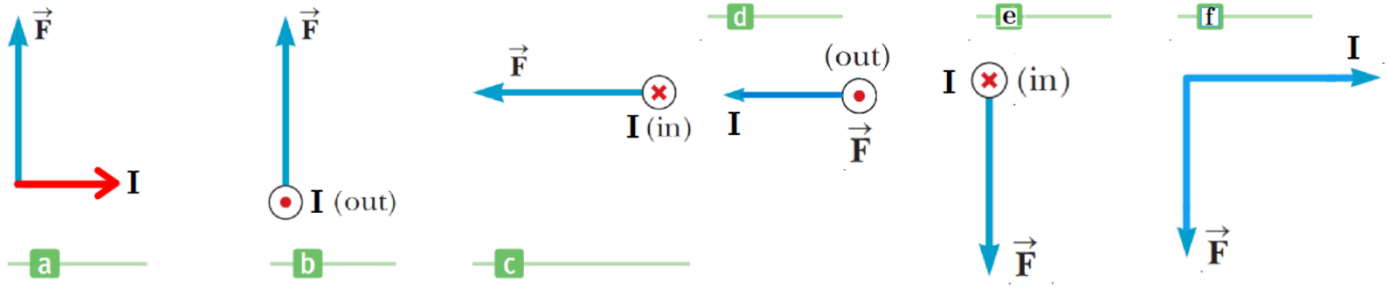
31. State the rule to determine the direction of a
(a) magnetic field produced around a straight conductor-carrying current,
(b) force experienced by a current-carrying straight conductor placed in a magnetic field which is perpendicular to it, and
(c) current induced in a coil due to its rotation in a magnetic field.

Lesson2 Magnetic Force (DC motor and AC generator)

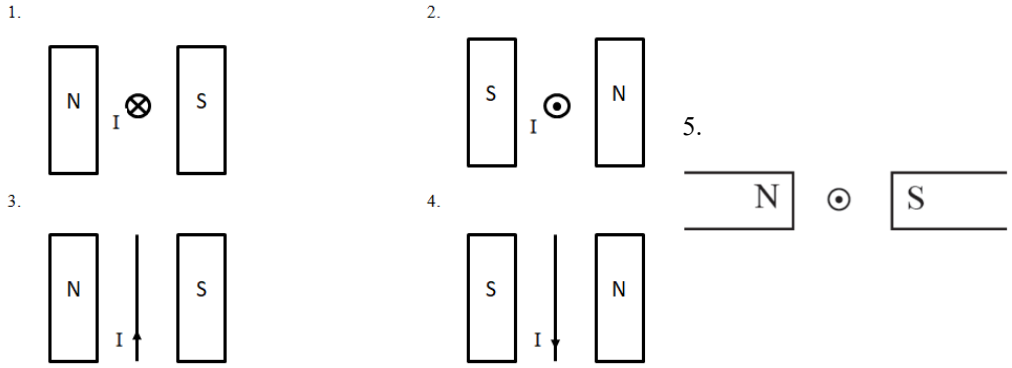
1. Use the answer key provided to indicate the direction of the force of the magnetic field on the current carrying wire in diagrams (a) to (h) below.



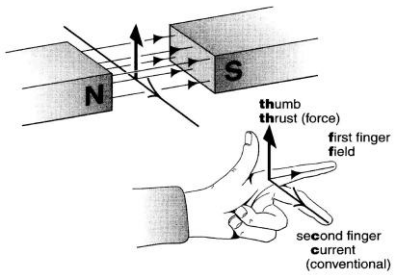
2. Find the direction of the magnetic field acting on a current-carrying wire in the various situations shown in Figure below if the direction of the magnetic force acting on it is as indicated.



3. Use Fleming's left-hand rule to find the direction of the magnetic force in each case below:



Fleming left-hand



4. **Figure 4.5.25** shows a diagram of an electric motor. As the current flows from the battery, the coil and split-ring commutator spin.

a. What is a simple d.c generator?

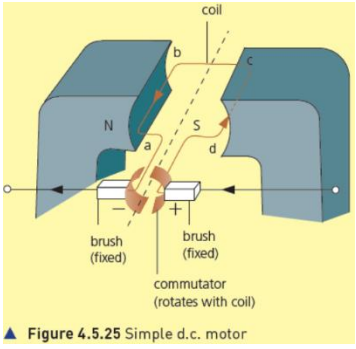
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b. A simple d.c. motor **(Fig 4.5.25)** contains a rectangular coil which is free to rotate between the poles of a magnet. Current flows into the coil via a pair of

…… and a ……(or split ring).



▲ **Figure 4.5.25** Simple d.c. motor

c. Indicate the direction of the magnetic force acting on sides **ab** and **cd.**

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d. What is the main purpose of a commutator or split ring and carbon brushes in a simple d.c motor?

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e. Suggest why side bc and ad did not experience a force.

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f. Explain why the coil spins when a current flows through it.

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g. When the coil is nearly vertical, the forces cannot turn it much further. Why does the coil keep spinning in the magnetic field even though there are no forces acting on both sides of it?

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h. Without changing the coil, give two ways in which it could be made to spin faster.

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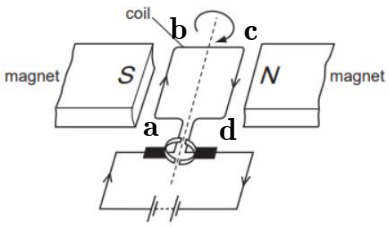
i. Give two ways in which the coil could be made to spin in the opposite direction.

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j. A rectangular loop of wire which can be made to rotate at a constant speed within a uniform magnetic field of $0.26\ \mathrm{T}$. The dimensions of the coil are $7\ \mathrm{cm} \times 4\ \mathrm{cm}$ and the current in loop is $2.0\ \mathrm{A}$. Find the magnetic force acting on the side **cd**, when side **cd** is



i) perpendicular to the magnetic field

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ii) parallel to the magnetic field.

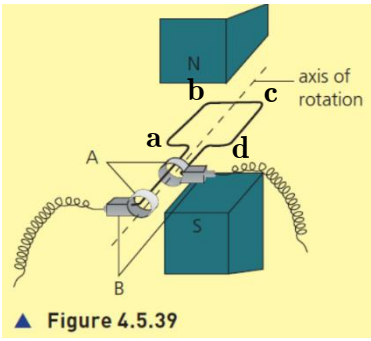
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5. A simple a.c. generator is shown in **Figure 4.5.39.**



▲ Figure 4.5.39

a. Name A and B and describe their purpose.

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b. What is an a.c generator?

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c. Describe changes that could be made to increase the e.m.f. generated.

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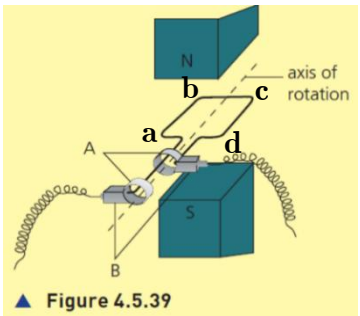
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d. Explain why no emf will ever be induced across **bc** and **ad**.

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e. Explain the direction of the induced current in the coil when it rotates clockwise in a magnetic field.



▲ Figure 4.5.39

f. Sketch a graph of e.m.f. against time for the generator and relate the position of the generator coil to the peaks, troughs and zeros of the e.m.f.

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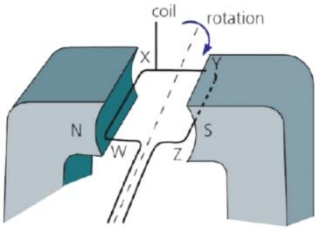
g. Explain why this apparatus can be described as an *ac* generator.

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h. A rectangular loop of wire which can be made to rotate at a constant speed within a uniform magnetic field of $0.26\ \mathrm{T}$. The dimensions of the coil are $6.2\ \mathrm{cm} \times 3.6\ \mathrm{cm}$ and the current in loop is $2.0\ \mathrm{A}$. Find the magnetic force acting on the side **YZ**, when side **YZ** is



i) perpendicular to the magnetic field

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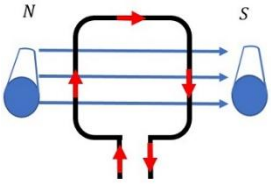
ii) parallel to the magnetic field.

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6. In the diagram, we are looking down on the setup. The current is moving clockwise around the coil. Which way does the coil rotate?



a. The right side goes down, and the left side goes up.
b. The right side goes up, and the left side goes down.
c. The whole coil moves up.
d. The whole coil moves down.

7. The device used for producing electric current is called a
(a) generator.
(b) galvanometer.
(c) ammeter.
(d) motor.

8. The essential difference between an AC generator and a DC generator is that
(a) AC generator has an electromagnet while a DC generator has permanent magnet.
(b) DC generator will generate a higher voltage.
(c) AC generator will generate a higher voltage.
(d) AC generator has slip rings while the DC generator has a commutator.

9. At the time of short circuit, the current in the circuit
(a) reduces substantially
(b) does not change
(c) increases heavily
(d) vary continuously

10. State whether the following statements are true or false.
(a) An electric motor converts mechanical energy into electrical energy.
(b) An electric generator works on the principle of electromagnetic induction.
(c) The field at the centre of a long circular coil carrying current will be parallel straight lines.
(d) A wire with a green insulation is usually the live wire of an electric supply.

Chapter4 Lesson3 Electromagnet and Transformers

11. Choose the best answer based on your understanding of the topic as follow

1. The magnetic field produced by a straight current-carrying wire depends on:
a) The length of the wire. **b)** The distance from the wire.
c) The material of the wire. **d)** The temperature of the wire.

2. The direction of the magnetic field lines around a straight wire is determined by:
a) Lenz's law. **b)** Ohm's law.
c) Faraday's law. **d)** The right-hand rule.

3. According to Ampere's law, the magnetic field produced by a straight current-carrying wire:
a) Is inversely proportional to the current. **b)** Is directly proportional to the current.
c) Is inversely proportional to the length of the wire. **d)** Is directly proportional to the radius of the wire.

4. The strength of the magnetic field due to a straight wire decrease:
a) Linearly with the distance from the wire. **b)** Exponentially with the distance from the wire.
c) Inversely proportional to the square of the distance from the wire.
d) Inversely proportional to the cube of the distance from the wire.

5. The mathematical expression for the magnetic field produced by a straight wire, at a distance r from the wire, is given by the equation:
a) $B = \mu_0 I/d$ **b)** $B = \mu_0 I/2\pi d$ **c)** $B = \mu_0 I/4\pi d^2$ **d)** $B = \mu_0 I/2\pi d^2$

6. A DC motor converts electrical energy into:
a) Mechanical energy. **b)** Heat energy. **c)** Light energy. **d)** Chemical energy.

7. In a DC motor, the component that carries the current and experiences the force in the presence of a magnetic field is called the:
a) Stator. **b)** Rotor. **c)** Armature. **d)** Commutator.

8. The magnetic field in a DC motor is typically created by:
a) Permanent magnets. **b)** Electromagnets. **c)** Both permanent magnets and electromagnets.
d) None of the above.

9. In an AC generator, mechanical energy is converted into:
a) Electrical energy. **b)** Thermal energy. **c)** Potential energy. **d)** Kinetic energy.

10. The rotating component of an AC generator that induces an alternating current in the wire is called the:
a) Stator. **b)** Rotor. **c)** Armature. **d)** Commutator.

11. The magnetic field in an AC generator is typically created by:
a) Permanent magnets. **b)** Electromagnets. **c)** Both permanent magnets and electromagnets.
d) None of the above.

12. As the armature of an AC generator rotates, the changing magnetic field induces a(n):
a) Direct current. **b)** Alternating current. **c)** Magnetic field. **d)** Electric field.

13. A solenoid is a coil of wire that produces a magnetic field when:
a) An electric current flow through it. **b)** It is placed near a magnet.
c) It is heated to a high temperature. **d)** It is exposed to sunlight.

14. The magnetic field inside a solenoid is:
a) Weak and non-uniform. **b)** Strong and non-uniform.
c) Weak and uniform. **d)** Strong and uniform.

15. The primary coil of a power transformer is connected to the:
a) Input voltage source. **b)** Output voltage source. **c)** Ground. **d)** Load.

16. The core of a power transformer is typically made of a _____ material to enhance the magnetic coupling between the coils.
a) Ceramic. **b)** Plastic. **c)** Ferromagnetic. **d)** Non-conductive.

17. The magnetic field in a power transformer is created by the _____ current in the primary coil.
a) AC (alternating current). **b)** DC (direct current). **c)** High-frequency current.
d) Low-frequency current.

18. A step-up transformer is designed to:
a) Increase the voltage. **b)** Decrease the voltage. **c)** Increase the current. **d)** Decrease the current.

19. Electrical power transformers are commonly used in:
a) Electronic devices. **b)** Lighting fixtures. **c)** Power transmission and distribution systems.
d) Battery charging systems.

20. The strength of the magnetic field inside a solenoid depends on:
a) The length of the solenoid. **b)** The number of turns in the solenoid.
c) The material of the solenoid. **d)** All of the above.

21. The magnetic field outside a solenoid is similar to that of a:
a) Strong permanent magnet. **b)** Weak permanent magnet.
c) Uniform electric field. **d)** Non-uniform electric field.

22. An electromagnet is created by:
a) Placing a permanent magnet inside a solenoid. **b)** Passing an electric current through a solenoid with ferromagnetic core.
c) Heating a solenoid to a high temperature. **d)** Cooling a solenoid to a low temperature.

12. (Choose the appropriate words from the box to fill in the blank as follows.

1. Straight wire	2. permeability	3. Solenoid	4. straight	5. magnitude
6. Electromagnet	7. decrease	8. closely	9. bar magnet	10. turns

1. A _____ is a long, cylindrical coil of wire that is often wound tightly to create a strong magnetic field when an electric current flow through it.
2. The magnetic field lines inside a solenoid is _____ and uniform along its length.
3. The strength of the magnetic field inside a solenoid depends on the _____ of the current, the number of _____ turns, and the permeability of the core material.
4. The magnetic field outside a solenoid is similar to that of a _____.
5. An _____ is a device that uses an electric current to generate a magnetic field. It typically consists of a solenoid with a ferromagnetic core.
6. The strength of the magnetic field in an electromagnet can be increased by increasing the number of _____, increasing the current, or using a core material with high _____.

1. mechanical	2. stationary, rotating	3. coil, permanent	4. alternating
5. electromagnets	6. coil	6. carbon brushes	7. direct

7. A DC motor operates based on the interaction between a _____ magnetic field and a _____ magnetic field.
8. In a DC motor, a current-carrying _____ is placed within a magnetic field created by _____ magnets.
9. A DC motor converts electrical energy into _____ energy through the interaction of magnetic fields.
10. In an AC generator, a _____ is rotated within a magnetic field, which induces an _____ current in the coil.
11. The output of an AC generator is typically an alternating current, which can be used to power various devices or be converted to a _____ current as needed.

1. Magnetic field	2. right-hand rule	3. Fleming left hand rule	4. current	5. distance
6. Electromagnet	7. decrease	8. $\mu_0 I$	9. $\mu_0 NI$	10. IlB

12. The _____ is a fundamental concept in electromagnetism and describes the influence of electric currents on the surrounding space.

13. The direction of the magnetic field lines around a straight wire can be determined using the _____rule, which states that if you point your thumb in the direction of the current flow, the curling of your fingers represents the direction of the magnetic field lines.

14. The strength of the magnetic field decreases with _____ from the wire. This relationship is inversely proportional to the distance.

15. The mathematical expression for the magnetic field produced by a straight wire, at a distance r from the wire, is given by the equation $B = \ldots\ldots / 2\pi d$, where I is the current flowing through the wire.

1. Voltage	2. Magnitude	3. Alternating	4. Ferromagnetic	
5. Electromagnets	6. Output	6. Input	7. Electrical	8. Long distances

16. An electrical power transformer is a device that is used to transfer _____ energy between two or more electrical circuits through electromagnetic induction.

17. The primary coil, also known as the _____ winding, is connected to the input voltage source.

18. The secondary coil, also known as the _____ winding, is connected to the load or the output circuit.

19. The core of a power transformer is typically made of a _____ material, such as iron or steel, to enhance the magnetic coupling between the coils.

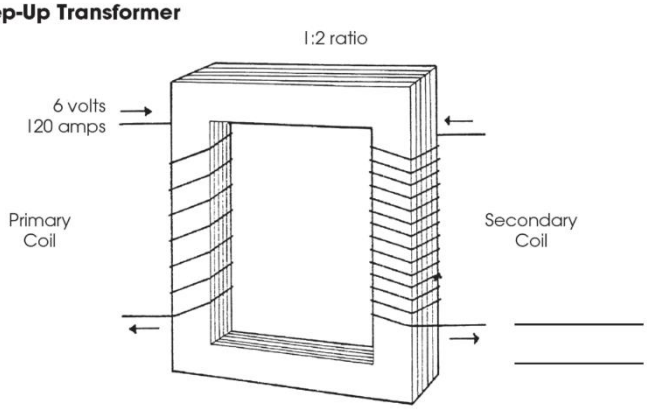
20. The primary and secondary coils are wound around the core, and the magnetic field created by the _____ current in the primary coil induces a voltage in the secondary coil.

21. A step-up transformer is designed to increase the _____ of the input voltage, while a step-down transformer is designed to decrease it.

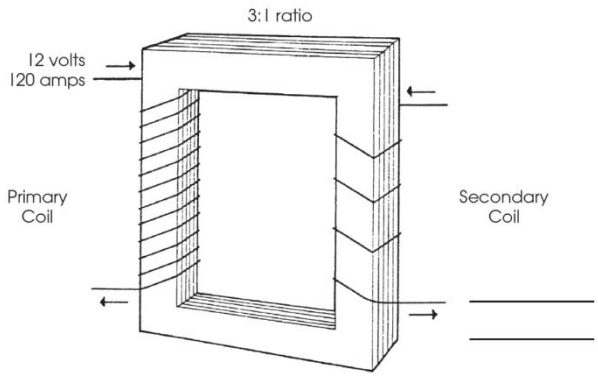
22. The ratio of the number of turns in the primary coil to the number of turns in the secondary coil determines the _____ ratio of the transformer.

13. **1.** Determine the voltage and current in each transformer.

Step-Up Transformer



Step-Down Transformer



2. A transformer has 600 primary turns connected to a 1.5 kV supply. Determine the number of secondary turns for a 240 V output voltage, assuming no losses.

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3. An ideal transformer with a turn ratio 2:9 is fed from a 220 V supply. Determine its output voltage.

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4. A transformer has 800 primary turns and 2000 secondary turns. If the primary voltage is 160 V, determine the secondary voltage assuming an ideal transformer.

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5. An ideal transformer with a turn ratio 3:8 has an output voltage of 640 V. Determine its input voltage.

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6. An ideal transformer has a turns ratio of 12:1 and is supplied at 192 V. Calculate the secondary voltage.

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7. A transformer primary winding connected across a 415 V supply has 750 turns. Determine how many turns must be wound on the secondary side if an output of 1.66 kV is required.

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8. An ideal transformer has a turns ratio of 15:1 and is supplied at 180 V when the primary current is 4 A. Calculate the secondary voltage and current.

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9. A step-down transformer having a turns ratio of 20:1 has a primary voltage of 4 kV and a load of 10 kW. Neglecting losses, calculate the value of the secondary current.

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10. A transformer has a primary to secondary turns ratio of 1:15. Calculate the primary voltage necessary to supply a 240 V load. If the load current is 3 A determine the primary current. Neglect any losses.

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11. A 10 kVA, single-phase transformer has a turns ratio of 12:1 and is supplied from a 2.4 kV supply. Neglecting losses, determine **(a)** the full load secondary current, **(b)** the minimum value of load resistance which can be connected across the secondary winding without the kVA rating being exceeded, and **(c)** the primary current.

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12. A 20 Ω resistance is connected across the secondary winding of a single-phase power transformer whose secondary voltage is 150 V. Calculate the primary voltage and the turns ratio if the supply current is 5 A, neglecting losses.

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13. A step-up transformer has a primary coil consisting of 200 turns and a secondary coil consisting of 3000 turns. The primary coil is supplied with an effective AC voltage of 90.0 V.

a. What is the voltage in the secondary circuit?

b. The current in the secondary circuit is 2.0 A. What is the current in the primary circuit?

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14. A step-down transformer has 7500 turns on its primary coil and 125 turns on its secondary coil. The voltage across the primary circuit is 7.2 kV. What voltage is being applied across the secondary circuit? If the current in the secondary circuit is 36 A, what is the current in the primary circuit?

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15. A step-up transformer has 300 turns on its primary coil and 90,000 turns on its secondary coil. The EMF of the generator to which the primary circuit is attached is 60.0 V. What is the EMF in the secondary circuit? The current in the secondary circuit is 0.50 A. What current is in the primary circuit?

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